

Task 6 : Create a Strong Password and Evaluate Its Strength

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Tools Used: <https://passwordmeter.com/>

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Objective

- To understand what makes a password strong and test it against password strength tools

Requirements for a good password

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Password Used : It"s0SiMpLe

Complexity and Security(A Formula to evaluate)

Password **complexity** and **length** directly translate to cryptographic strength, a metric known as **Entropy** (measured in bits).

$$\text{Entropy} = L * \log_2(C)$$

Where **L** is the **Length** of the password, and **C** is the size of the **Character Set** used.

Observation

Test Your Password		Minimum Requirements
Password:	<input s0simple"="" type="text" value="It"/>	- Minimum 8 characters in length - Contains 3/4 of the following items: - Uppercase Letters - Lowercase Letters - Numbers - Symbols
Hide:	☐	
Score:	94%	
Complexity:	Very Strong	

Additions		Type	Rate	Count	Bonus
	Number of Characters	Flat	$+(n*4)$	11	+ 44
	Uppercase Letters	Cond/ Incr	$+\left((len-n)*2\right)$	4	+ 14
	Lowercase Letters	Cond/ Incr	$+\left((len-n)*2\right)$	5	+ 12
	Numbers	Cond	$+(n*4)$	1	+ 4
	Symbols	Flat	$+(n*6)$	1	+ 6
	Middle Numbers or Symbols	Flat	$+(n*2)$	2	+ 4
	Requirements	Flat	$+(n*2)$	5	+ 10
Deductions					
	Letters Only	Flat	$-n$	0	0
	Numbers Only	Flat	$-n$	0	0
	Repeat Characters (Case Insensitive)	Comp	-	0	0
	Consecutive Uppercase Letters	Flat	$-(n*2)$	0	0

Conclusion

This task underscored that the objective of strong password creation is to make the computational effort required for guessing or cracking the password **exceed the economic and time limits** of any attacker.